

# A Programmable Address Space for Neural Memory

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## Abstract

Neural networks store knowledge in distributed parameters but do not normally expose an operational address space whose contents can be selectively written, reassigned and transported while declared responses are preserved. We assign the restricted response kernel a new operational role: not merely as a local source of response-preserving directions, but as the carrier from which a finite memory-address structure can be instantiated. We introduce Moving Fibre Intelligence, a framework that tests this role using ridge-regularized near-kernel updates for operations within a response-constrained parameter set and local response retraction for transport between declared response sets.

In separately frozen audits of pretrained GPT-2 using restricted LoRA-B parameter charts, writable endpoints supported exact retrieval, overwrite and access through linguistic expressions excluded from within-run writing updates. Two binary slots formed four compositional codes that were reused under cyclic concept-code assignments. An eight-endpoint atlas spanning 28 pairwise relations was subsequently transported to a distinct response set and returned while preserving exact and held-out-expression access. Mean relative transport distortion was  $1.45 \times 10^{-5}$ , compared with  $7.89 \times 10^{-5}$  for matched-scale random-kernel controls, and direct retraction produced lower distortion in all 28 dependent pairwise comparisons.

We then tested whether these resources could operate together in one shared parameter state. Four address cells were jointly written, one was selectively overwritten, two exchanged their contents, and the resulting state was moved to a second response set and returned. Every stage preserved exact and fresh held-out-expression access, positive read margins and the declared response and KL budgets; where applicable, unselected cells retained their declared values. Overwrite and swap also outperformed wrong-operation and matched-norm random-update controls.

Within the tested GPT-2 LoRA charts, these results demonstrate an operational role for the restricted response kernel as the local carrier of a finite compositional address geometry, with numerically near-kernel updates supporting within-set operations and response retraction supporting local transport between sets. The procedures are externally specified and gradient-based, and the evidence comes from separate single-seed protocols. Autonomous instruction execution, unified multi-seed confirmation, capacity scaling, cross-model replication and global transport remain open.

## 1 Introduction

Representation answers what internal state correlates with a concept; addressing asks whether a concept can be assigned to and retrieved from explicit operational locations using reusable address codes. Attention, key-value caches, sparse routing and external memory all implement addressing over activations or architecture. The question here concerns parameters: whether a response-constrained region of parameter space can carry an explicit, auditable address code, whether its instantiated structure survives local transport, and whether these resources can support selective operations on one shared parameter state.

Four traditions bear on this problem: distributed representation [2, 4], associative and key-value memory [5, 12], model editing [8–10], and parameter symmetries and flat directions [1, 3, 6]. Within these traditions, null directions have commonly appeared as redundancy or non-identifiability, and in some learning and editing methods as constraints for preserving selected behaviour. What remains missing is their organization into an explicit address geometry whose codes can be composed, reassigned and operated on in a shared parameter state.

The central mathematical shift of this paper is therefore not to discover that null directions can preserve behaviour, but to test whether a restricted response kernel can serve as the local carrier of an explicit and

operable memory-address geometry. Concretely, we take

$$\mathcal{V}_\theta = \ker(DR_{A,S}(\theta)|_{T_\theta \mathcal{L}}) \quad (1)$$

as that carrier in its ideal first-order form. Ridge-regularized near-kernel updates instantiate finite address states within a declared response-constrained set, while local response retraction transports those instantiated states between distinct declared sets. This kernel–retraction decomposition provides the geometric basis for the compositional, remappable and shared-state operations tested below.

Operationally, we declare a finite response map on a fixed anchor set, take the parameters that keep that response within two declared budgets, and write each concept–code pair to its own endpoint in that set; Section 2 gives the definitions. Three construction steps then make the address atlas. Writing and overwriting at a controlled endpoint establishes that these locations hold memories. Access through expressions excluded from the writing updates establishes that a read is not a recital of a training string. Cyclic reassignment of two-slot codes across concept identities establishes that a code is not the name of its concept. These three are prerequisites, not parallel results.

These construction steps establish the resources required for an address atlas, but not yet their transportability or shared-state operation. In a separate prospectively frozen transport audit, we therefore ask whether an instantiated atlas preserves its readout and relational geometry between distinct response-constrained parameter sets. Applying one local response-retraction rule separately to each endpoint of an eight-endpoint cohort carried the cohort from one finite-response set to another while preserving every read and, to within a declared aggregate gate, all 28 pairwise relations among the endpoints.

The final test is operational. We ask whether the preceding capabilities can be realized together in one shared state by executing a frozen sequence of joint WRITE, selective OVERWRITE, SWAP, MOVE and RETURN. OVERWRITE and SWAP are additionally audited against wrong-operation and matched-norm random-update controls. This shared-state audit is the culminating experiment of the paper.

Together, these audits form a staged mechanism chain from reusable address construction, through local transport, to selective shared-state operation within one restricted GPT-2 LoRA chart.

## 2 Operational address geometry

### 2.1 Declared response map and admissible set

Let  $S$  denote the prospectively fixed selector of the declared response coordinates. For an anchor set  $A = \{x_1, \dots, x_m\}$ , fixed before evaluation, the audited response map is

$$R_{A,S}(\theta) = \frac{1}{|A|} \sum_{x \in A} S f_\theta(x). \quad (2)$$

Because the selector  $S$  is fixed throughout each declared protocol, we suppress it in the notations  $\mathcal{F}_A$ ,  $\mathcal{Y}_A$ ,  $\mathcal{F}_s$  and  $\mathcal{F}_t$ . Response distances are measured in the supremum norm on the selected coordinates,  $d_R(y, y') = \|y - y'\|_\infty$ , so a response budget bounds the largest single-coordinate deviation rather than an aggregate. For a declared target response  $r$  and reference state  $\theta_{\text{ref}}$ , the finite admissible set under the response and collateral-KL budgets is

$$\mathcal{F}_A(r, \varepsilon) = \{\theta : d_R(R_{A,S}(\theta), r) \leq \varepsilon_R, \quad \bar{D}_{\text{KL},A}(\theta_{\text{ref}}, \theta) \leq \varepsilon_{\text{KL}}\}, \quad (3)$$

where the collateral budget is the anchor-averaged divergence

$$\bar{D}_{\text{KL},A}(\theta_{\text{ref}}, \theta) = \frac{1}{|A|} \sum_{x \in A} D_{\text{KL}}(p_{\theta_{\text{ref}}}(\cdot | x) \| p_\theta(\cdot | x)). \quad (4)$$

The KL reference is fixed prospectively within each protocol. In the staged shared-state audits it is the initial pre-program state  $\theta_{\text{base}}$  and is not reset between operations; in the endpoint-wise transport audit it is

the corresponding pre-transport endpoint  $\theta_{c,a}^{(s)}$ . The source chart is recovered by taking  $r_s = R_{A,S}(\theta_{\text{ref}})$ ; a target chart uses a separately declared  $r_t$ , so that

$$\mathcal{F}_s := \mathcal{F}_A(r_s, \epsilon), \quad \mathcal{F}_t := \mathcal{F}_A(r_t, \epsilon). \quad (5)$$

These are finite budgeted response sets, not assumed to form a smooth global fibre bundle.

Let  $\mathcal{L} \subset \mathbb{R}^p$  denote the restricted LoRA-B parameter chart, the only coordinates updated in this work, and let  $\mathcal{B}_A$  denote the finite-dimensional response-coordinate space of the declared anchor map. At  $\theta \in \mathcal{L}$  the response differential is  $DR_{A,S}(\theta) : T_\theta \mathcal{L} \rightarrow \mathcal{B}_A$ , and the local response kernel is

$$\mathcal{V}_\theta := \ker(DR_{A,S}(\theta)|_{T_\theta \mathcal{L}}) = \{v \in T_\theta \mathcal{L} : DR_{A,S}(\theta)v = 0\}, \quad (6)$$

so that  $R_{A,S}(\theta + \eta v) = R_{A,S}(\theta) + o(\eta)$  as  $\eta \rightarrow 0$  for  $v \in \mathcal{V}_\theta$ ; no claim is made about the kernel of the full GPT-2 parameter space.

In practice the kernel is realized numerically. With  $J_\theta = DR_{A,S}(\theta)|_{T_\theta \mathcal{L}}$ , the ridge-regularized near-kernel operator used throughout is

$$P_{\theta,\lambda} = I - J_\theta^\top (J_\theta J_\theta^\top + \lambda I)^{-1} J_\theta, \quad \lambda = 10^{-8}. \quad (7)$$

We call  $P_{\theta,\lambda}$  a numerical near-kernel operator rather than an exact projector, because  $\lambda > 0$  generally implies  $P_{\theta,\lambda}^2 \neq P_{\theta,\lambda}$ . Candidate tangent updates take the form  $v = P_{\theta,\lambda}u$  and lie near, rather than exactly in,  $\mathcal{V}_\theta$ ; they are accepted only after the declared finite response and KL gates are evaluated.

## 2.2 Concept, code, endpoint, readout

Let  $\mathcal{C}$  denote the finite concept cohort and let

$$A := \{0, 1\}^2 \quad (8)$$

be the four-element address-code set. An assignment is a declared map  $\pi : \mathcal{C} \rightarrow A$ . Writing concept  $c$  under assignment  $\pi$  produces an endpoint  $\theta_{c,\pi(c)}$  with auditable readout  $\sigma(c, \pi(c))$ , and the endpoint atlas instantiated by  $\pi$  is

$$\mathcal{E}_\pi := \{\theta_{c,\pi(c)} : c \in \mathcal{C}\} \subset \mathcal{F}_s. \quad (9)$$

The load-bearing relation is therefore

$$(c, \pi(c)) \mapsto \theta_{c,\pi(c)} \mapsto \sigma(c, \pi(c)). \quad (10)$$

The same code may occur at several concept-specific endpoints, so an address code does not identify a unique global parameter point. A slot is one coordinate of the code, and slots are audited separately.

**Response kernel as a local carrier.** Three levels should be kept apart:  $\mathcal{V}_\theta$  is the ideal first-order local carrier,  $P_{\theta,\lambda}u$  is the numerically projected near-kernel candidate direction, and  $\text{Retract}(\theta + \eta P_{\theta,\lambda}u) \in \mathcal{F}_A(r, \epsilon)$  is the finite state that the declared budgets actually admit. Accordingly,  $\mathcal{V}_\theta$  is the local carrier of the tested address operations; it is neither the discrete code set  $A$  nor the instantiated endpoint atlas  $\mathcal{E}_\pi$ .

## 2.3 Shared state and operation family

A shared memory state  $\Theta_m \in \mathcal{F}_s$  realizes a content map

$$m : A \rightarrow \{0, 1\} \quad (11)$$

simultaneously within one parameter state. This object is distinct from the endpoint atlas  $\mathcal{E}_\pi$ , whose elements are separately written concept-code endpoints. With  $a, b \in A$ , the target state relations audited in Section 5 are

$$\text{WRITE}(\Theta, m) \mapsto \Theta_m, \quad \text{OVERWRITE}_a(\Theta_m, v) \mapsto \Theta_{m[a \leftarrow v]}, \quad (12)$$

$$\text{SWAP}_{a,b}(\Theta_m) \mapsto \Theta_{m^{a \leftrightarrow b}}, \quad \text{MOVE}_{s \rightarrow t}(\Theta_m^{(s)}) \mapsto \Theta_m^{(t)}, \quad (13)$$

$$\text{RETURN}_{t \rightarrow s}(\Theta_m^{(t)}) \mapsto \widehat{\Theta}_m^{(s)}. \quad (14)$$

Here  $m^{a \leftrightarrow b}$  exchanges the values assigned to cells  $a$  and  $b$  and leaves every other cell unchanged, and  $m[a \leftarrow v]$  sets cell  $a$  to  $v$  with the other cells untouched, while  $\Theta_m^{(s)} \in \mathcal{F}_s$  and  $\Theta_m^{(t)} \in \mathcal{F}_t$  realize the same declared content map  $m$  in different response-constrained sets. These are the target state relations of the experimental procedures, not an algebra proved closed over arbitrary states, and not an instruction language parsed by the model. Whether any of them can be executed while the declared budgets hold is the question of Section 5.

## 2.4 Local response retraction and transport

Candidate within-set updates are selected by backtracking and followed by ridge-regularized response correction; a proposal with no admissible step is a zero step. The correction iterates the following ridge-regularized minimum-norm update for the linearized response equation:

$$\delta\theta = -J_\theta^\top (J_\theta J_\theta^\top + \lambda I)^{-1} (R_{A,S}(\theta) - r), \quad \lambda = 10^{-8}, \quad (15)$$

applied for a declared number of iterations and stopped early once the response residual falls below  $10^{-7}$ . It is the regularized response-correcting counterpart associated with the same Jacobian: the near-kernel operator suppresses first-order response change, whereas this update removes the remaining linearized response residual. MOVE and RETURN apply these declared iterations directly and are accepted only when their post hoc gates hold. Applied endpoint by endpoint, that retraction carries endpoints between finite-response sets,

$$\mathcal{I}_{s \rightarrow t} : \mathcal{E}_\pi^{(s)} \longrightarrow \mathcal{E}_\pi^{(t)}, \quad \mathcal{I}_{s \rightarrow t}(\theta_{c,a}^{(s)}) = \theta_{c,a}^{(t)}, \quad (16)$$

where  $\mathcal{E}_\pi^{(s)} \subset \mathcal{F}_s$  and  $\mathcal{E}_\pi^{(t)} \subset \mathcal{F}_t$ . The notation names an experimentally specified endpoint-wise retraction map on a finite audited cohort, not a global connection or parallel transport.

## 2.5 Protocol provenance

The evidence comes from separately frozen protocols rather than one continuous or joint multi-seed experiment, and no statistic is pooled across stages. L1 is a single-seed development record; the archived standalone L2 protocol lacks its original numerical result; and the L3, L4 and L5 records are separate prospective single-seed confirmation candidates. Held-out-expression claims are drawn from the archived L3–L5 records, not reconstructed from the missing L2 result. Protocol identifiers, seeds, hashes, complete gate definitions and evidence locations are reported in the corresponding evidence directories and repository manifests. A subsequent post-failure development smoke combined representative capabilities from all stages in one continuous shared-state sequence and passed all declared smoke gates; because its configuration was adapted after earlier failures on the same seed, it is reported only as joint-realizability development evidence and does not replace the separately frozen stage-specific records.

# 3 Constructing an address atlas

## 3.1 Writable, readable and overwritable endpoints

A declared value was written at a controlled endpoint and read back exactly; a second value was then written at the same location and read in place of the first; both endpoints satisfied the declared budgets. Random-kernel, sign-reversed and no-move arms bound the alternative that any sufficiently small parameter change would produce the same reads. Only endpoints meeting both budgets enter any positive claim.

## 3.2 Access through excluded expressions

Expressions excluded from the within-run gradient updates and from the checkpoint-selection rule read the same written values. The positive held-out-expression claim reported here is supported by the archived compositional-addressing and transport records, both of which prospectively evaluate expressions excluded

from their within-run writing updates; the standalone expression-access development protocol is retained as historical provenance but is not used as an independently archived numerical result. Because the protocols were produced through adaptive development on the same experimental family, this remains single-seed evidence rather than an independently replicated generalization test.

### 3.3 Assignment reuse and current-assignment readout

The address-code set is  $A = \{0, 1\}^2$ , represented as  $\{00, 01, 10, 11\}$  and formed from two separately audited slots, with a mixed-slot minimum-margin gate so that a code using both slots must be read with positive margin rather than on average. Let  $\pi_j : c \mapsto a$  be the  $j$ -th cyclic assignment. Each assignment is written independently,  $(c, \pi_j(c)) \mapsto \theta_{c, \pi_j(c)}^{(j)}$ , so the four assignments occupy sixteen separately initialized endpoints, and the audited property is

$$\text{Read}(\theta_{c, \pi_j(c)}^{(j)}) = \pi_j(c). \quad (17)$$

The code rule is reused across concept identities and readout follows the current assignment.

| $j$ | $\pi_j$                              | Exact | Held-out | Mixed-slot minimum margin |
|-----|--------------------------------------|-------|----------|---------------------------|
| 0   | dog 00; Tokyo 01; Japan 10; apple 11 | 8/8   | 32/32    | 1.9989                    |
| 1   | dog 01; Tokyo 10; Japan 11; apple 00 | 8/8   | 32/32    | 2.3205                    |
| 2   | dog 10; Tokyo 11; Japan 00; apple 01 | 8/8   | 32/32    | 1.6406                    |
| 3   | dog 11; Tokyo 00; Japan 01; apple 10 | 8/8   | 32/32    | 1.8391                    |

Across the sixteen endpoints every exact and held-out read was correct, the minimum mixed-slot training margin was 1.6406 against a gate of 0.5, and every endpoint stayed inside the declared response and KL budgets. The result constructs a reusable address atlas, but each concept–code pair still occupies a separately initialized endpoint. A training-only semantic router, correct on 79 of 80 outer-held-out concepts, assigned all four bridge instances to their categories and hence, through a fixed category-to-address map, to their codes. Per-view counts, the individual routing margins and the pair-only and shuffled-label control values are in the archived records.

*This is assignment reuse on independently initialized endpoints, not in-place rebinding of a single endpoint and not rejection of a former binding. Shared-state storage of four cells, and selective operations on them, are audited separately in Section 5. The archived record labels this run a prospective single-seed confirmation candidate requiring multi-seed confirmation.*

## 4 Transport across response fibres

### 4.1 Cohort and operation

With  $\mathcal{F}_s$  and  $\mathcal{F}_t$  as declared in Section 2.1, the source cohort is the endpoint atlas

$$\mathcal{E}_\pi^{(s)} = \{\theta_{c, \pi(c)}^{(s)} : c \in \mathcal{C}, \pi(c) \in A\} \subset \mathcal{F}_s, \quad |\mathcal{C}| = 8. \quad (18)$$

The cohort holds eight concept-specific endpoints over four repeated two-slot codes, two concepts per code: the eight endpoints are not eight distinct addresses, and the code set is  $A$ , not a three-slot cube. The retraction map of Section 2.4 is applied endpoint by endpoint, so the operation is endpoint-level while the audit is cohort-level.

### 4.2 Declared audited quantities

Three quantities are audited: the readout  $I_{\text{read}} = \text{Read}(\theta_{c,a})$ , target-response eligibility

$$I_{\text{elig}}(\theta; r) = \mathbf{1}[d_R(R_{A,S}(\theta), r) \leq \varepsilon_R \wedge \bar{D}_{\text{KL},A}(\theta_{c,a}^{(s)}, \theta) \leq \varepsilon_{\text{KL}}], \quad (19)$$

where the collateral budget of the transport stage is recorded relative to that endpoint’s own pre-transport state rather than to a global base state, and the pairwise geometry  $I_{\text{pair}} = \{d(\theta_{c,a}, \theta_{c',a'})\}$ . The response value itself is not an invariant: the target set is displaced from the source by a declared shift of 0.08, so what is preserved is eligibility for the set the endpoints are carried to, not  $R_{A,S}(\theta_{c,a})$ . Geometric preservation is measured by the mean relative distortion over the  $|\mathcal{P}| = 28$  unordered endpoint pairs,

$$D_{\text{geom}} = \frac{1}{|\mathcal{P}|} \sum_{(u,v) \in \mathcal{P}} \frac{|d_t(\theta_u^{(t)}, \theta_v^{(t)}) - d_s(\theta_u^{(s)}, \theta_v^{(s)})|}{d_s(\theta_u^{(s)}, \theta_v^{(s)}) + \varepsilon}, \quad u = (c, a), \quad v = (c', a'). \quad (20)$$

### 4.3 Random-kernel perturbation control

For each endpoint the control first adds a numerically projected response-kernel direction,

$$\theta^{\text{ctrl}} = \text{Retract}_{r_t}(\theta^{(s)} + z), \quad z = P_{\theta, \lambda} \xi, \quad \|z\| = \|\theta^{(t)} - \theta^{(s)}\|, \quad (21)$$

and then applies the same retraction rule to reach the target response set. The control matches the norm of the added kernel perturbation, not the final displacement, so the comparison asks whether direct retraction preserves cohort geometry better than retraction preceded by a matched-scale random tangent perturbation.

### 4.4 Result

| Cohort                          | Value     | Geometry                               | Value                  |
|---------------------------------|-----------|----------------------------------------|------------------------|
| Concept-specific endpoints      | 8         | Transport distortion $D_{\text{geom}}$ | $1.451 \times 10^{-5}$ |
| Unique two-slot codes           | 4         | Control median distortion              | $7.888 \times 10^{-5}$ |
| Endpoint-pair relations         | 28        | Distortion ratio                       | 0.184                  |
| Response displacement           | 0.08      | Descriptive pairwise wins              | 28/28                  |
| Exact access after transport    | preserved | Largest single-pair error              | $3.97 \times 10^{-5}$  |
| Held-out access after transport | preserved | Round-trip distortion                  | $1.313 \times 10^{-7}$ |

Every exact and held-out read survived transport; the mean distortion over the 28 endpoint-pair relations met the declared aggregate gate, with a largest single-pair relative error of  $3.97 \times 10^{-5}$ ; and composing transport with its reverse returned each endpoint to itself,  $\mathcal{T}_{t \rightarrow s} \circ \mathcal{T}_{s \rightarrow t}(\theta_{c,a}) \approx \theta_{c,a}$ , at a round-trip distortion of  $1.313 \times 10^{-7}$ . Direct retraction produced lower distortion than the control on all 28 pairs. Transport shows that an instantiated atlas can move without losing its declared structure; it does not yet show that several address cells can coexist and be selectively manipulated in one state.

*Two qualifications travel with these numbers. The declared gate is aggregate, not per-pair: it applies to  $D_{\text{geom}}$  rather than to each of the 28 relations separately. And the 28 comparisons share the eight endpoints, so they are not independent Bernoulli trials: the exact sign-test value of  $p = 3.73 \times 10^{-9}$  that they would imply under independence was a declared gate of the frozen protocol but is reported here as a protocol diagnostic only, never as a significance result, in line with the statistical note archived with the run. The symbol  $\approx$  above is defined by the declared response, read and pair-geometry gates, and closure over one source–target pair does not imply path independence.*

### 4.5 Alternative explanations and transport controls

The result is not explained by independent endpoint refitting alone, because the audit requires preservation of all pairwise relations in addition to per-endpoint eligibility and readout. Nor is it explained by step size: a matched-scale random kernel perturbation followed by the same retraction produced approximately five times greater mean distortion. Round-trip recovery to  $1.313 \times 10^{-7}$  further excludes a strongly lossy projection on the tested cohort. These controls distinguish local structure-preserving transport from ordinary reassignment or unconstrained small movement, while remaining limited to one source–target pair. The remaining question is operational: can write, overwrite, rearrangement and transport be composed on one shared address state?

## 5 Programmable operations on a shared address state

The preceding sections establish writable locations, expression-level access, reusable compositional codes and locally transportable geometry. We now audit whether the target state relations defined in Section 2.3 can be realized as selective operations within one shared parameter state while satisfying the declared response and KL budgets. The audit is a prospectively frozen single-seed confirmation candidate at seed 82931, run with a fresh seed and fresh held-out expressions after adaptive development of the protocol family.

### 5.1 Program stages

Four binary cells are written jointly into a single shared state, one cell is then selectively overwritten, and two cells exchange their contents. WRITE, OVERWRITE and SWAP are response-constrained projected-gradient updates on the restricted LoRA coordinates; the base model stays frozen. Every stage is scored on exact reads and on fresh held-out expressions, and must leave the declared values of unselected cells unchanged.

| Stage             | Declared bits | Cells changed | Unselected preserved | Response drift        | Anchor KL |
|-------------------|---------------|---------------|----------------------|-----------------------|-----------|
| WRITE             | 0011          | —             | —                    | $1.53 \times 10^{-5}$ | 0.00374   |
| OVERWRITE at 01   | 0111          | {1}           | yes                  | 0                     | 0.00351   |
| SWAP of 00 and 11 | 1110          | {0,3}         | yes                  | $7.63 \times 10^{-6}$ | 0.00399   |

The observed bits matched the declared bits at every stage, exact and held-out accuracy were 1.0 throughout, and the smallest held-out margin over the three stages was 0.6154, at the initial write.

### 5.2 Operation controls

OVERWRITE and SWAP are each accompanied by two controls: a wrong operation, which executes a different declared state transition, and a matched-norm random-kernel update of the same update magnitude. On the requested-cell evaluation the overwrite reached exact and held-out accuracy 1.0, against 0.5 for the wrong instruction and 0.75 for the matched-norm random update at an update norm of 10.0006. The swap likewise reached 1.0, against 0.5 for both its wrong-instruction and matched-random controls at an update norm of 13.2654. The requested operation is therefore not reproduced by executing some other declared operation, nor by a displacement of the same size in the response kernel.

### 5.3 Moving the shared state

The shared four-cell state was then carried to a second response set and returned. The parameter displacement was 0.0978, the target-response residual  $1.53 \times 10^{-5}$  and the return residual  $1.53 \times 10^{-5}$ . Exact and held-out accuracy were 1.0 both after transport and after return, and the positive held-out margin gate held at both ends. Thus all five externally specified operations were successfully executed in the declared sequence on the shared four-cell state under the frozen access, selectivity, response and KL gates, and all sixteen declared gates of the audit passed.

*The operation names label externally executed experimental procedures, not instructions parsed by the model. WRITE, OVERWRITE and SWAP are response-constrained projected-gradient updates; MOVE and RETURN are local response retractions. The audit is a single-seed confirmation candidate: it does not establish multi-seed or model-family confirmation, autonomous instruction execution, gradient-free or inference-time memory operations, a general instruction-set architecture, arbitrary program composition, long-horizon error bounds, concept-level capacity scaling, in-place semantic rebinding beyond the declared binary content operations, or global parallel transport.*

## 6 Discussion

### 6.1 A discrete address system inside a continuous chart

The experiments identify the response kernel not only as a set of response-preserving null directions, but as a local carrier on which addressable memory operations can be realized. They therefore distinguish the carrier  $\mathcal{V}_\theta$ , the finite admissible set  $\mathcal{F}_A(r, \varepsilon)$ , the endpoint atlas  $\mathcal{E}_\pi$  and the shared state  $\Theta_m$ : these objects play related but non-interchangeable roles. The first structural result is that a continuous constrained parameter chart supports a finite discrete code system whose slots can be audited separately and recombined. Concepts are instantiated at endpoints  $\theta_{c,a}$ , the code is read from separately audited slots, and every positive claim is an audit of declared finite quantities rather than an inferred capacity of the model at large.

### 6.2 From concept binding to code reuse

The second result is separation: the same compositional codes can serve different concept identities, so the tested code is not merely another name for one concept. Readout follows the declared assignment at each separately written endpoint, and the mixed-slot margin gate rules out a writer that resolves one slot and guesses the other.

### 6.3 From geometric object to operational object

The shared-state result changes the role of the geometry. In the construction stages the address structure is built and read; under transport it is moved; in the shared state it becomes the state space of selective operations. The progression is from location, to code, to transport, and finally to programmability, and each stage consumes what the previous one produced.

This is also what separates the study from model editing. The object is not a single factual modification but the selective execution and composition of several operations on one finite address state, and neither control reproduced the requested state transition: the effect is tied to operation identity rather than displacement magnitude alone.

## 7 Limitations

- Every stage uses one pretrained GPT-2 model, one restricted LoRA-B chart and single-seed evidence; multi-seed and cross-model confirmation are absent.
- L1–L4 use separately initialized concept–code endpoints. Shared-state storage is tested only for four binary cells in Section 5, not for scaled concept-level capacity or in-place semantic rebinding.
- The operations are externally executed gradient-based procedures, not model-parsed or inference-time instructions. Arbitrary programs, long-horizon error bounds and a general instruction-set architecture are untested.
- L4 uses four repeated two-slot codes over eight endpoints and applies transport endpoint by endpoint between one source–target pair at one response displacement; it does not define a global connection or parallel transport.
- The random control matches the added kernel perturbation rather than the final displacement, and the 28 pairwise comparisons share eight endpoints; the declared geometry gate is aggregate rather than inferential.
- Router bridge instances come from its fitting cohort, the category-to-address map is hand-declared, and address information enters via a controlled prompt interface rather than emerging autonomously.
- The standalone L2 result archive is unavailable. No numerical result is reconstructed from prose; held-out-expression claims rely on the publicly archived later-stage records.



## 8 Conclusion

We have traced a staged path from writable parameter-space locations to a finite programmable address space. Exact anchoring and held-out-expression access establish readable locations; compositional re-assignment separates address code from concept identity; local transport preserves the instantiated atlas across response-constrained sets; and a shared-state audit combines these resources into selective WRITE, OVERWRITE, SWAP, MOVE and RETURN operations. Within one prospectively evaluated GPT-2 LoRA seed, all declared access, selectivity, response, KL and control gates of that audit passed. The result is a mechanism candidate for programmable neural memory, not an autonomous instruction set or a general neural computer. Multi-seed replication, cross-model validation, capacity scaling, longer programs and instruction-conditioned execution remain open.

## Data and code availability

Code, frozen configurations, result records and claim boundaries are available at <https://github.com/papasop/fibre-nav> under the archived L1–L5 evidence directories. This manuscript audits repository commit c243b0da8b8a05a2ebaffa6d310f8ca33b0416d7.

## References

1. S. Ainsworth, J. Hayase, and S. Srinivasa. Git Re-Basin: merging models modulo permutation symmetries. *International Conference on Learning Representations*, 2023.
2. T. Bricken et al. Towards monosemanticity: decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
3. L. Dinh, R. Pascanu, S. Bengio, and Y. Bengio. Sharp minima can generalize for deep nets. *International Conference on Machine Learning*, PMLR 70:1019–1028, 2017.
4. N. Elhage et al. Toy models of superposition. *Transformer Circuits Thread*, 2022.
5. M. Geva, R. Schuster, J. Berant, and O. Levy. Transformer feed-forward layers are key-value memories. *Conference on Empirical Methods in Natural Language Processing*, 2021.
6. R. Hecht-Nielsen. On the algebraic structure of feedforward network weight spaces. In *Advanced Neural Computers*, pages 129–135. Elsevier, 1990.
7. E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.
8. K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 2022.
9. K. Meng, A. Sharma, A. Andonian, Y. Belinkov, and D. Bau. Mass editing memory in a transformer. *International Conference on Learning Representations*, 2023.
10. E. Mitchell, C. Lin, A. Bosselut, C. Finn, and C. D. Manning. Fast model editing at scale. *International Conference on Learning Representations*, 2022.
11. C. Olah et al. Zoom in: an introduction to circuits. *Distill*, 2020.
12. S. Sukhbaatar, A. Szlam, J. Weston, and R. Fergus. End-to-end memory networks. *Advances in Neural Information Processing Systems*, 2015.